

The AI Rally vs. Fama-French: What 1,000 Random Portfolios Reveal About Mega-Cap Dominance

Josh Carpenter | July 2026 | Monthly data 2012–2026, Kenneth French Data Library + Yahoo Finance

The puzzle

I ran the Fama–French three-factor model on an equal-weight portfolio of 15 large-cap U.S. stocks, estimating rolling 36-month regressions from 2012 to today. For a decade the model worked almost embarrassingly well: R^2 sat between 0.90 and 0.97 through the pre-COVID years, the COVID crash, and the 2022 tightening cycle. Then, as the AI-rally months entered the estimation window, it slid from **0.96 to 0.78**. Two stories could explain that decline:

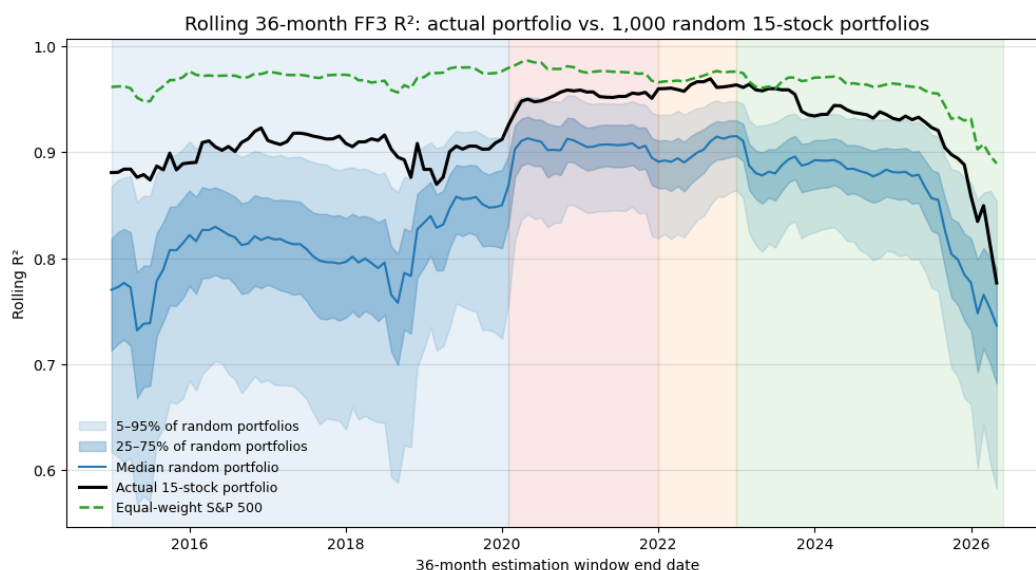
- **Stock-specific:** my particular AI winners (NVDA, MSFT, AAPL) became dominated by idiosyncratic moves — the model is fine, my 15 names aren't.
- **Market-wide:** the three-factor model genuinely explained a smaller share of equity returns during the AI era.

Why the obvious control fails

The natural instinct is to compare against SPY. But the market factor (Mkt–RF) is the value-weighted return of essentially the whole U.S. market, and the S&P 500 is roughly 85% of that capitalisation. Regressing SPY on the factors is nearly regressing a variable on itself: its rolling R^2 never leaves the 0.99–1.00 band in fourteen years of data (I checked). It *cannot* decline, so the comparison would force the “stock-specific” conclusion no matter what the truth is.

The experiment: 1,000 counterfactual portfolios

Instead, I downloaded all 503 current S&P 500 constituents (438 with full history since 2012), drew 1,000 random equal-weight portfolios of exactly 15 names, and ran the identical rolling regression on every one of them. That produces a sampling distribution (a “fan”) of R^2 paths for a portfolio of my size, and it controls for the mechanical fact that any 15-stock portfolio is noisier than an index. If my portfolio sank through the fan, my picks were the problem. If the whole fan sank, the market changed.



The verdict: the decline is market-wide

- **The whole fan fell as much as my portfolio.** From its post-2023 peak, my portfolio's R^2 dropped 0.19 (0.96 $\rightarrow$ 0.78); the median random portfolio dropped 0.18 (0.92 $\rightarrow$ 0.74).
- **My portfolio never entered the left tail.** It sits near the 95th percentile of the fan in every regime — including the AI rally. It rode the fan down; it didn't sink through it.
- **Even full diversification didn't escape.** The equal-weight S&P 500 — where no single stock's noise can matter — lost 0.09 of R^2 and now sits at its 14-year minimum.

Thus, the stronger claim survives: during the late AI rally, the three-factor model explained a materially smaller fraction of monthly returns for a wide range of portfolios drawn from today's S&P500 constituents.

What is the model not capturing?

Adding more factors answers part of the question. The five-factor model (profitability and investment, RMW + CMA) recovers about a third of the lost R^2 (0.78 $\rightarrow$ 0.85 in the latest window) — the AI winners are extreme high-profitability, aggressive-investment stocks, a dimension value (HML) doesn't span. Momentum adds nothing. The remaining two-thirds is structural:

- **Co-movement collapsed.** Average pairwise correlation among S&P 500 stocks fell from 0.43 (2022) to 0.24; cross-sectional return dispersion rose from 6.7% to 9.4% per month. When stocks move together less, any common-factor model explains less — there is simply less common variance to capture.
- **The market factor decoupled from the average stock.** The rolling correlation between the equal-weight S&P 500 and Mkt–RF fell from 0.99 to 0.87 — unprecedented in this sample.

This is what mega-cap dominance looks like in the data

That last number is the statistical fingerprint of a sentiment everyone has been repeating: **a handful of mega-caps are the market now.** Because the market factor is cap-weighted, its variance is increasingly driven by a few AI giants. “The market went up” and “the average stock went up” used to be the same statement. In the AI era they diverged. The benchmark itself has drifted away from the typical stock, so factor models calibrated on broad co-movement quietly lose their grip because the index it's anchored to no longer represents the cross-section it was meant to summarise. Mega-cap dominance isn't just an index-weight trivia point; in this sample, it shows up as measurable decline in how well broad factors explain the average large-cap stock.

Caveats

Universe is today's constituents backfilled to 2012 (the results are subject to survivorship bias which makes the exercise cleaner but less historically pure; an improved version would use point-in-time S&P500 membership). Monthly data, 36-month overlapping windows (estimates are serially dependent); R^2 comparisons are descriptive, not formal tests. Random draws are not sector-balanced, which is why a deliberately diversified portfolio sits high in the fan. Fixed seed; full code and notebook available on request.